PowerPulse

Real-Time Electricity Demand & Generation Analytics

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DATA-226

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Introduction

Every hour, the U.S. electric grid quietly pumps out an avalanche of data — demand spikes, generation drops, regional quirks, and those mysterious weekend patterns that nobody talks about.

Our project pulls **hourly operational data** from the **U.S. Energy Information Administration (EIA) Electric Power Operations dataset**, and runs it through a modern data pipeline powered by **Airflow, Snowflake, dbt, and Power BI** to transform this raw operational grid data into **actionable insights** that reveal demand patterns, regional behavior, and system trends across U.S. balancing authorities.

The result?
A fully automated ETL → ELT → Analytics workflow that turns raw grid data into **interactive dashboards, trend insights, and region-vs-region showdowns.**
We uncover how the grid behaves, when it misbehaves, and why Mondays always look different from Saturdays.

In short: we try to make electricity data easier and fun to explore.

The Challenge

Electricity grids operate in real-time with zero room for error. Operators must balance supply and demand instantly, yet traditional forecasting methods rely on manual processes that can't keep pace with volatile renewable generation and unpredictable consumption patterns.

- Electricity grids face volatile demand patterns
- Real-time generation must match consumption to prevent blackouts
- Traditional manual forecasting is slow and error-prone
- Need for automated, data-driven insights for grid operators

Data Sources

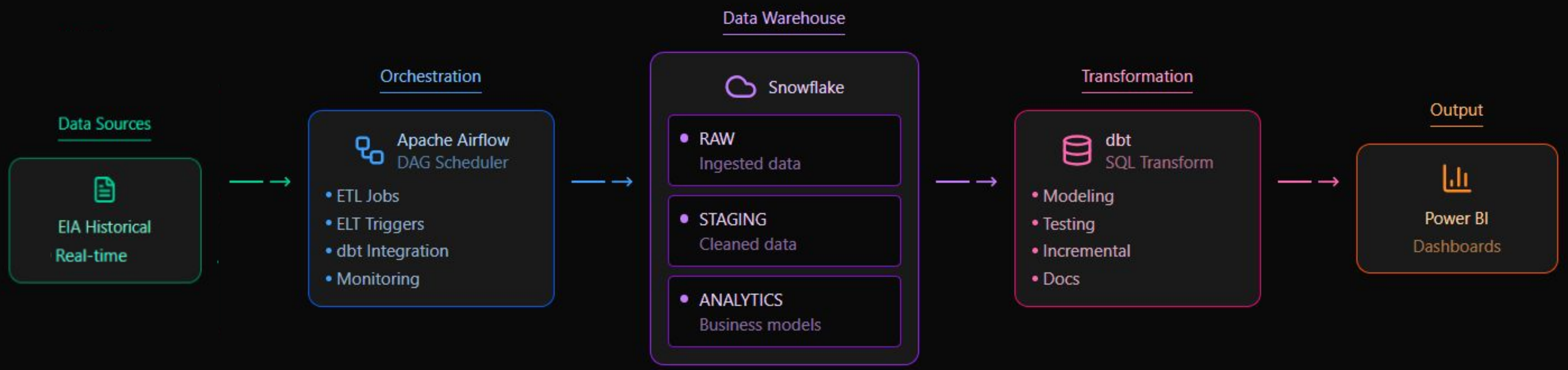
EIA Data

- **Source:** U.S. Energy Information Administration API
- **Coverage:** 7 days
- **Granularity:** Hourly demand data
- **Format:** JSON/XML feeds
- **Metrics:** MWh consumption, generation mix
- **Geography:** State and regional level

Combined: ~8M+ hourly records from 50+ data series

End-to-End Data Analytics Architecture

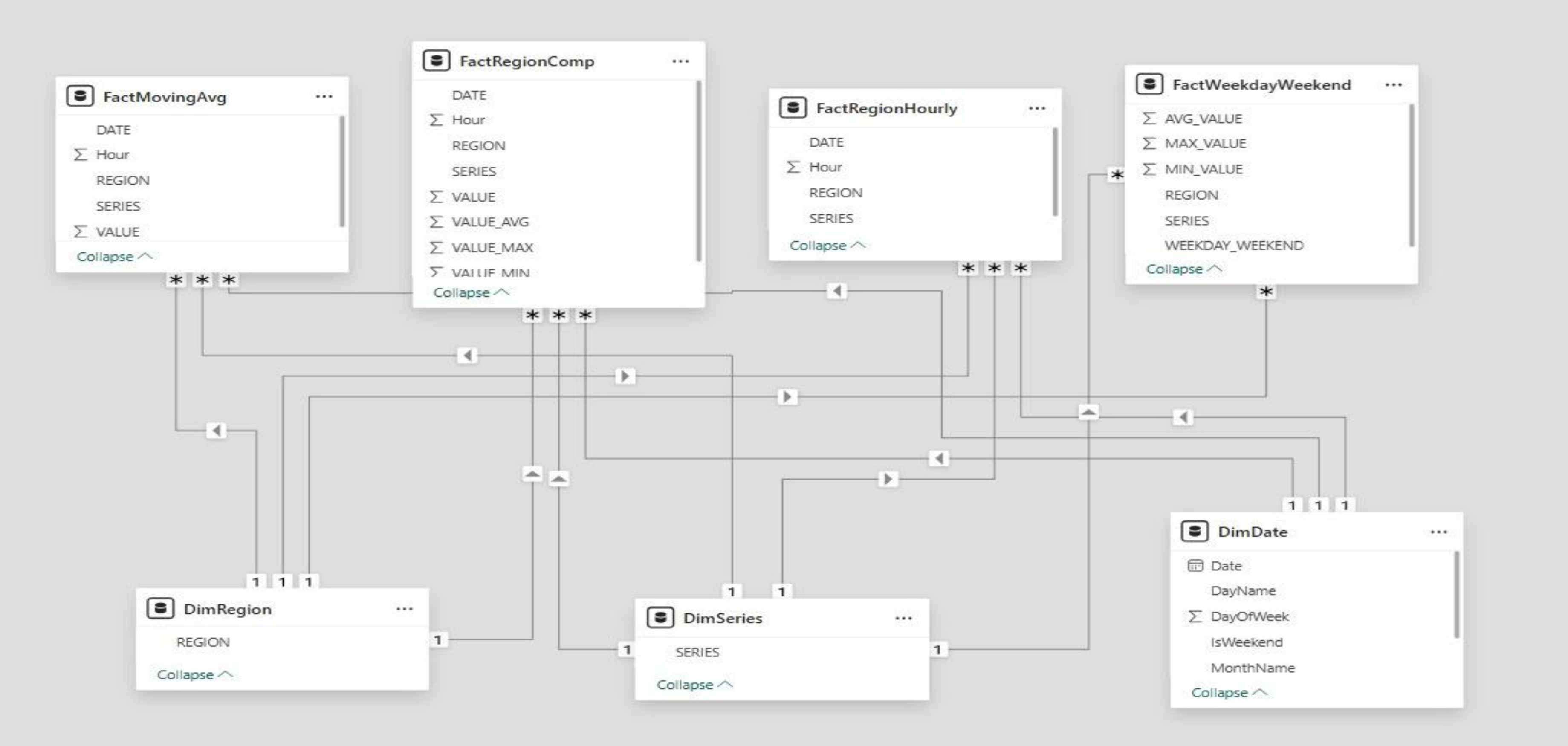
ETL + ELT Pipeline with Airflow & dbt Orchestration



Data Flow Summary

Historical electricity data is extracted from EIA Open Data API, orchestrated by Airflow pipelines, loaded into Snowflake's multi-layer architecture (raw → staging → analytics), transformed using dbt models to create analytical datasets, and visualized through interactive Power BI/Tableau dashboards for grid operators and decision makers to monitor demand, generation, and forecast accuracy.

Database Schema



ETL and ELT Process :

End-to-End Data Flow, Source → Ingestion → Snowflake

and

Transformation with dbt, integrated with Apache Airflow

ETL: Data Ingestion Flow

- **Data Extraction** 📥:
  We start by pulling hourly operational data straight from the U.S. Energy Information Administration **(EIA)** – Electric Power Operations dataset. Think of this as scooping up real-time snapshots of how the U.S. grid is behaving every single hour.

- **Airflow in Action** : Airflow DAG, “eia_ETL”, jumps into action on a set schedule and manages extraction from the EIA API.
  Raw responses are processed in Python to

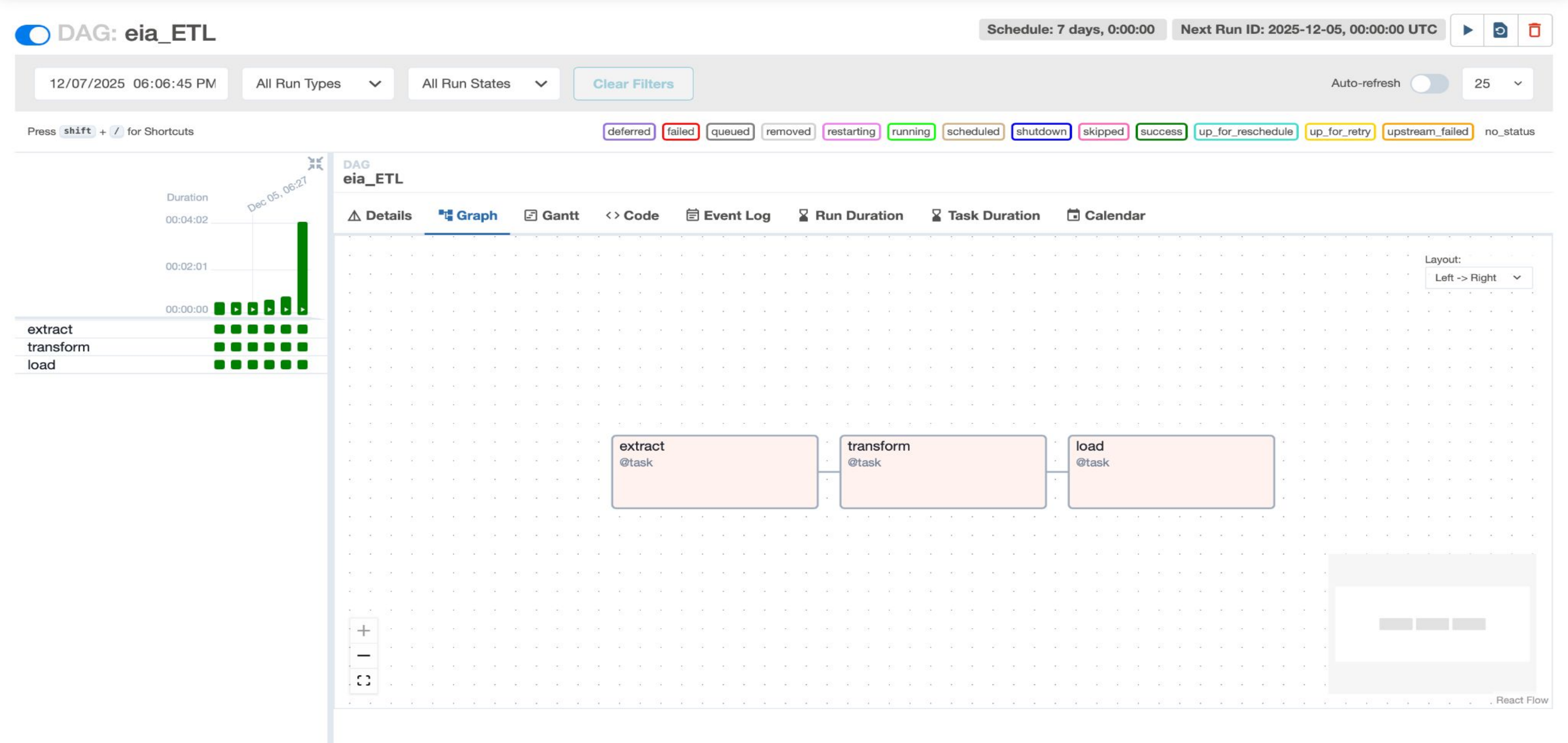
  Standardize column names and data types

  Convert timestamps to a unified format

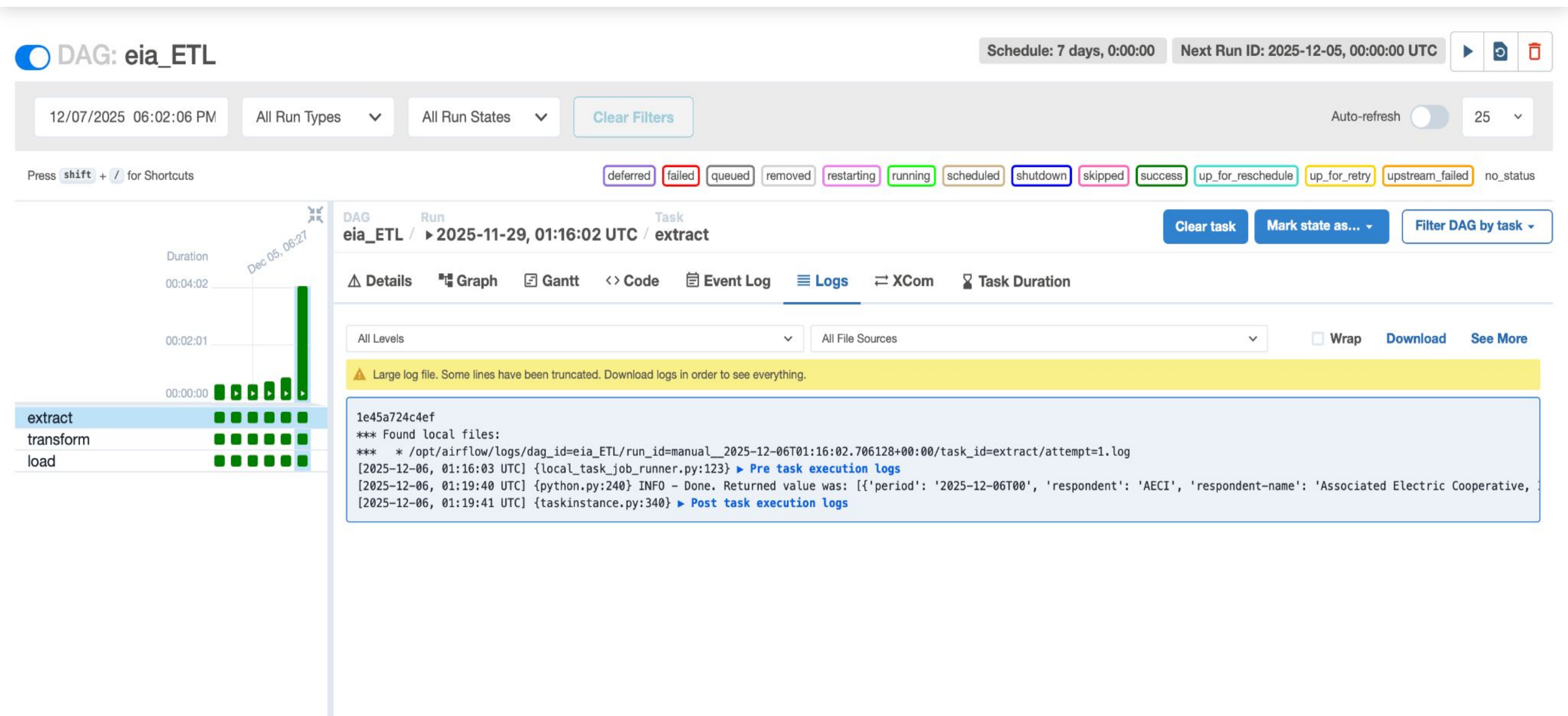
  Handle inconsistent values

- **Load into Snowflake** : The cleaned hourly dataset is loaded into Snowflake as “RTO_REGION_HOURLY”.
  This table serves as the single trusted landing layer, ensuring consistent inputs for downstream ELT transformations.

ETL:

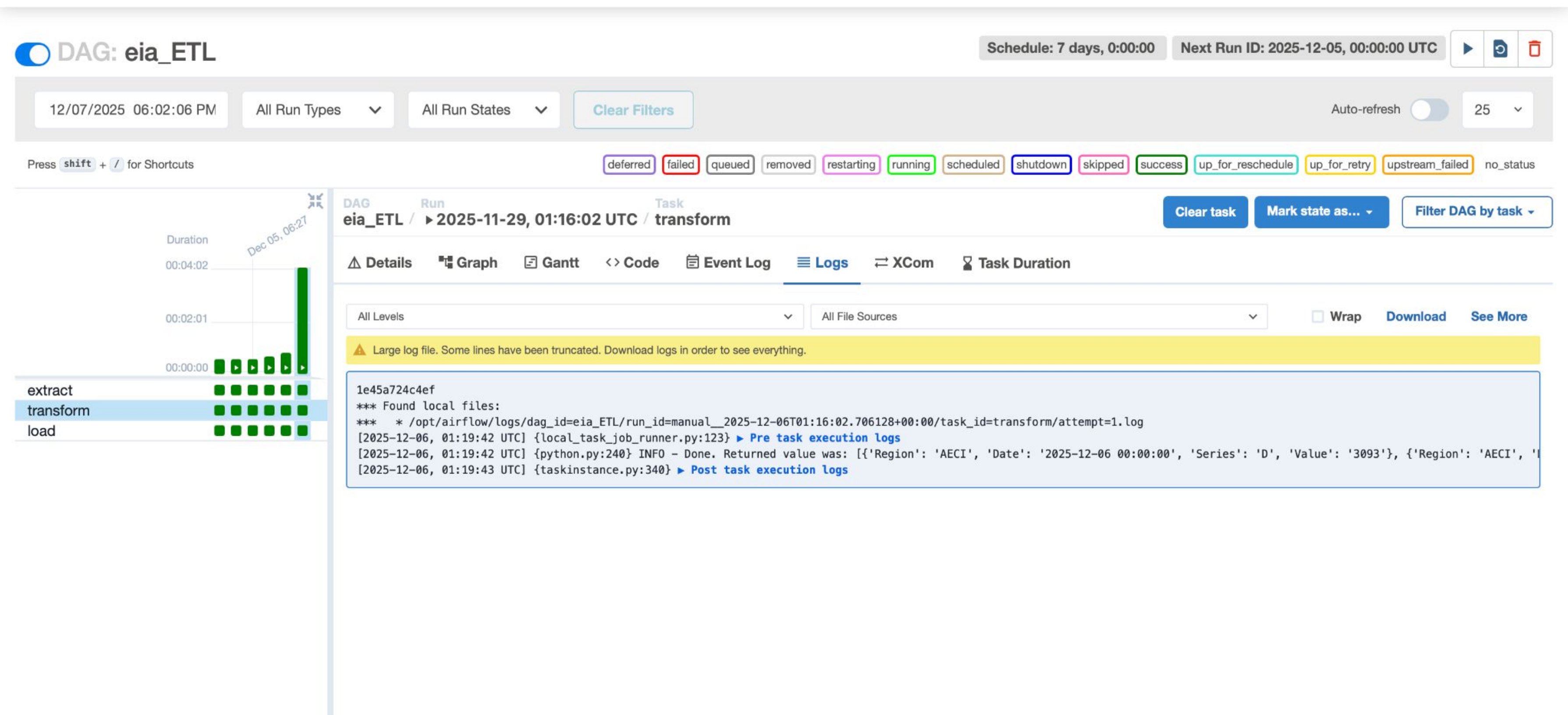

DAG: eia_ETL
Schedule: 7 days, 0:00:00
Next Run ID: 2025-12-05, 00:00:00 UTC
12/07/2025 06:06:45 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred
failed
queued
removed
restarting
running
scheduled
shutdown
skipped
success
up_for_reschedule
up_for_retry
upstream_failed
no_status
Duration
Dec 05, 06:27
00:04:02
00:02:01
00:00:00
extract
transform
load
DAG
eia_ETL
Details
Graph
Gantt
Code
Event Log
Run Duration
Task Duration
Calendar
Layout:
Left -> Right
extract
@task
transform
@task
load
@task
React Flow

ETL (extract):

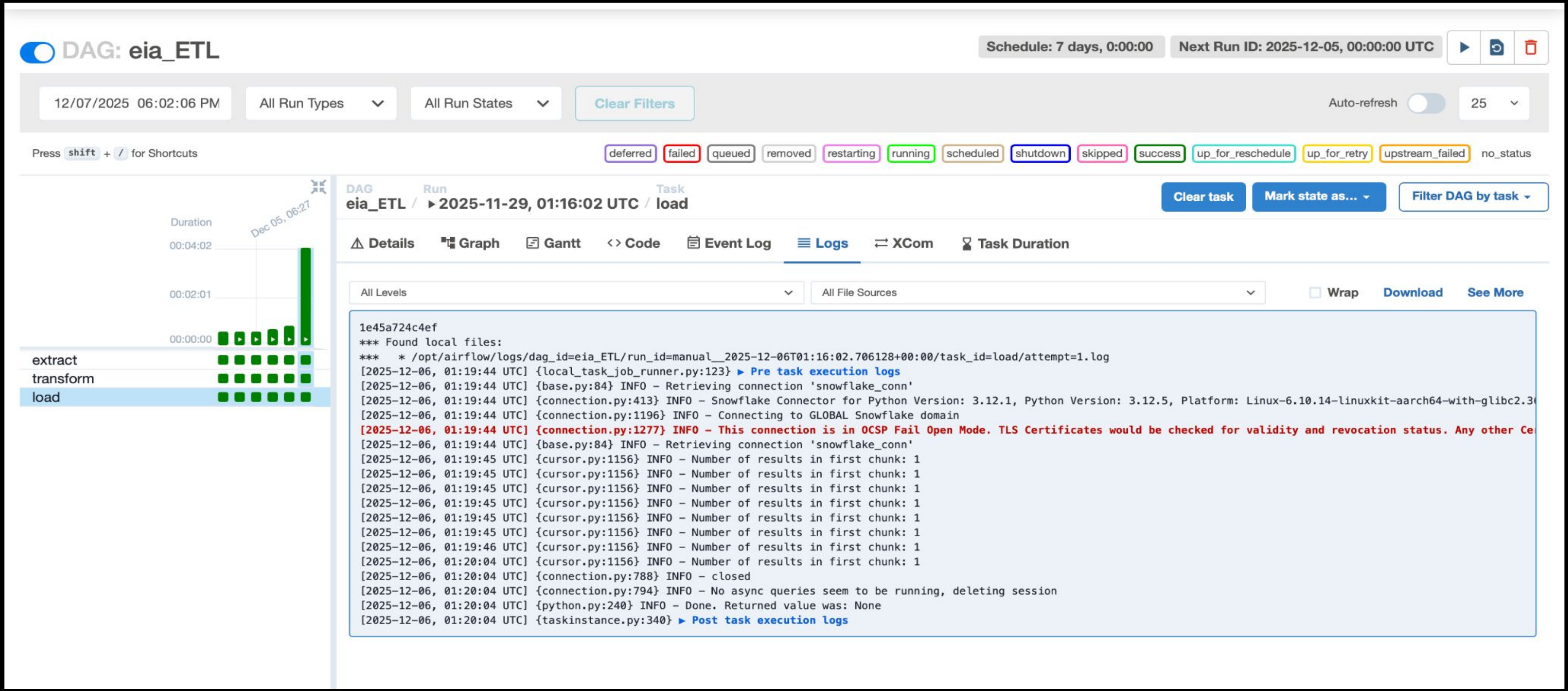


DAG: eia_ETL
Schedule: 7 days, 0:00:00
Next Run ID: 2025-12-05, 00:00:00 UTC
12/07/2025 06:02:06 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred failed queued removed restarting running scheduled shutdown skipped success up_for_reschedule up_for_retry upstream_failed no_status
Duration
Dec 05, 06:27
00:04:02
00:02:01
00:00:00
extract
transform
load
DAG eia_ETL / Run 2025-11-29, 01:16:02 UTC / Task extract
Clear task
Mark state as...
Filter DAG by task
Details Graph Gantt Code Event Log Logs XCom Task Duration
All Levels
All File Sources
Wrap
Download
See More
Large log file. Some lines have been truncated. Download logs in order to see everything.
1e45a724c4ef
*** Found local files:
*** * /opt/airflow/logs/dag_id=eia_ETL/run_id=manual__2025-12-06T01:16:02.706128+00:00/task_id=extract/attempt=1.log
[2025-12-06, 01:16:03 UTC] {local_task_job_runner.py:123} ▶ Pre task execution logs
[2025-12-06, 01:19:40 UTC] {python.py:240} INFO - Done. Returned value was: [{'period': '2025-12-06T00', 'respondent': 'AECI', 'respondent-name': 'Associated Electric Cooperative,
[2025-12-06, 01:19:41 UTC] {taskinstance.py:340} ▶ Post task execution logs

ETL (transform):



ETL (load):


DAG: eia_ETL
Schedule: 7 days, 0:00:00
Next Run ID: 2025-12-05, 00:00:00 UTC
12/07/2025 06:02:06 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred failed queued removed restarting running scheduled shutdown skipped success up_for_reschedule up_for_retry upstream_failed no_status
Duration
Dec 05, 06:27
00:04:02
00:02:01
00:00:00
extract
transform
load
DAG eia_ETL / Run 2025-11-29, 01:16:02 UTC / Task load
Clear task
Mark state as...
Filter DAG by task
Details Graph Gantt Code Event Log Logs XCom Task Duration
All Levels
All File Sources
Wrap
Download
See More
1e45a724c4ef
*** Found local files:
*** * /opt/airflow/logs/dag_id=eia_ETL/run_id=manual__2025-12-06T01:16:02.706128+00:00/task_id=load/attempt=1.log
[2025-12-06, 01:19:44 UTC] {local_task_job_runner.py:123} ▶ Pre task execution logs
[2025-12-06, 01:19:44 UTC] {base.py:84} INFO - Retrieving connection 'snowflake_conn'
[2025-12-06, 01:19:44 UTC] {connection.py:413} INFO - Snowflake Connector for Python Version: 3.12.1, Python Version: 3.12.5, Platform: Linux-6.10.14-linuxkit-aarch64-with-glibc2.3
[2025-12-06, 01:19:44 UTC] {connection.py:1196} INFO - Connecting to GLOBAL Snowflake domain
[2025-12-06, 01:19:44 UTC] {connection.py:1277} INFO - This connection is in OCSP Fail Open Mode. TLS Certificates would be checked for validity and revocation status. Any other Ce
[2025-12-06, 01:19:44 UTC] {base.py:84} INFO - Retrieving connection 'snowflake_conn'
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:45 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:19:46 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:20:04 UTC] {cursor.py:1156} INFO - Number of results in first chunk: 1
[2025-12-06, 01:20:04 UTC] {connection.py:788} INFO - closed
[2025-12-06, 01:20:04 UTC] {connection.py:794} INFO - No async queries seem to be running, deleting session
[2025-12-06, 01:20:04 UTC] {python.py:240} INFO - Done. Returned value was: None
[2025-12-06, 01:20:04 UTC] {taskinstance.py:340} ▶ Post task execution logs

ELT + Analytics

- **ELT Transformation with dbt (integrated with airflow):** The **eia_ELT_dbt DAG** executes dbt workflows directly inside Snowflake, ensuring scalable SQL-based transformations.

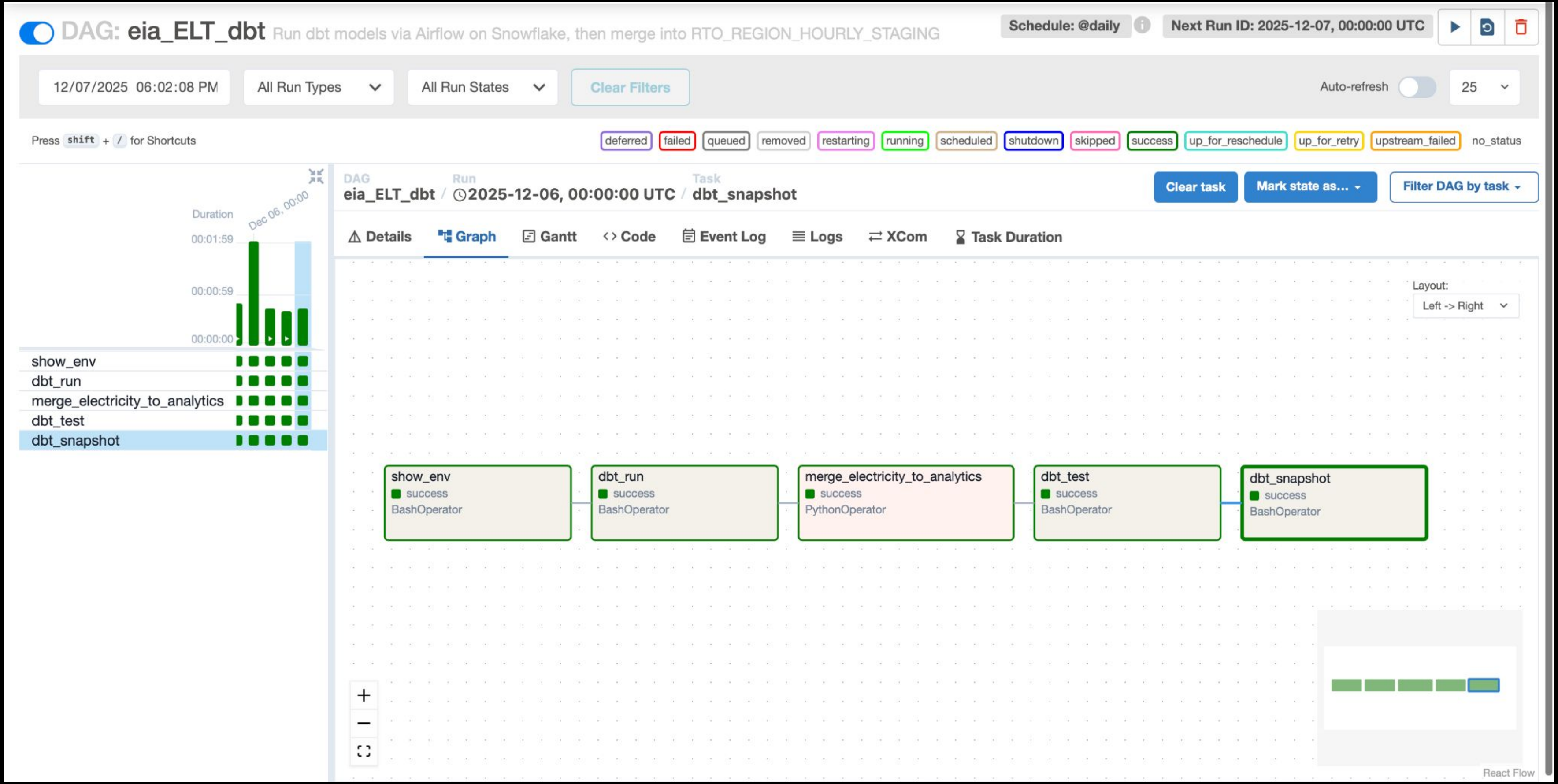
- **dbt steps executed:**
  dbt run → builds staging + analytics models
  dbt test → validates data quality (null checks, accepted value)
  dbt snapshot → tracks historical changes for slowly evolving attributes and auditability

- **Core dbt Models:**
  stg_eia – standardized base table (REGION, DATE, SERIES, VALUE).
  region_comp – computes daily min/max/avg + region ranking via SQL window functions.
  moving_avg – calculates 7-day moving averages to highlight long-term trends.
  weekday_weekend – aggregates usage patterns across weekday vs weekend behavior.

- The resulting **Analytics Layer** in Snowflake provides fast, structured data for real-time insights.

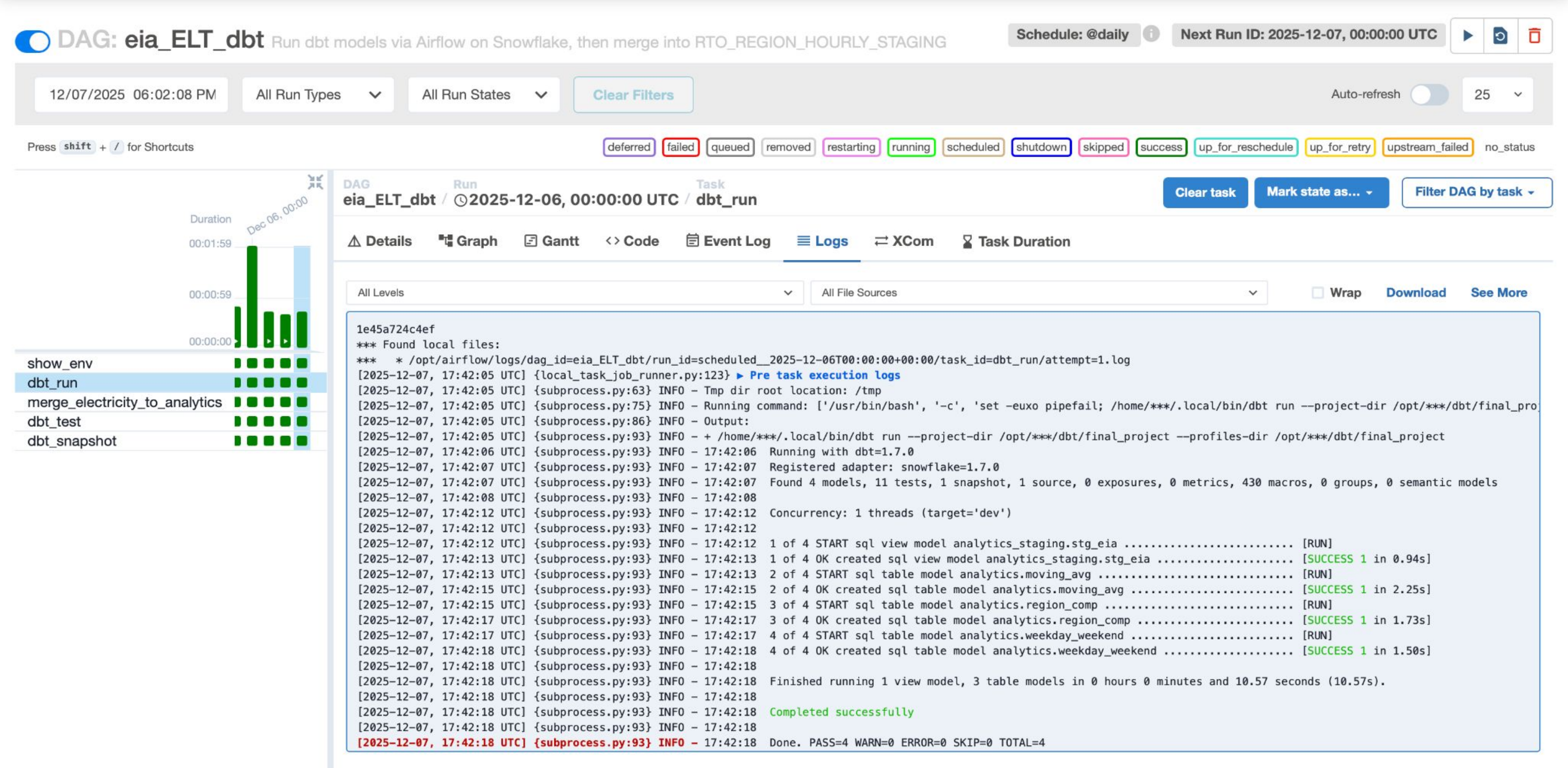
- **Visualization in Power BI** :
  Final dbt models feed directly into Power BI dashboard, enabling exploration of trends, regional comparisons, and consumption behavior with interactive visuals.

ELT:

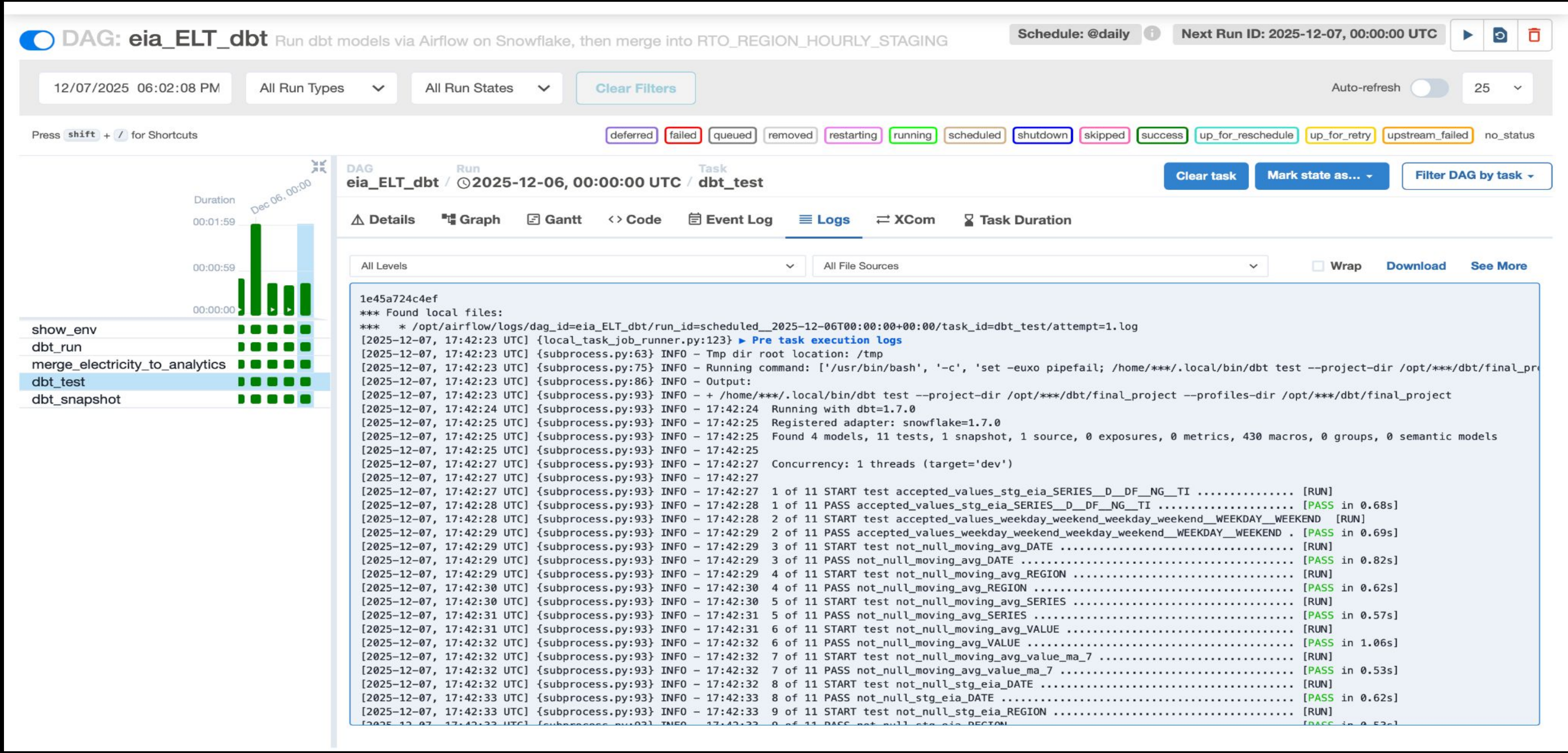


DAG: eia_ELT_dbt Run dbt models via Airflow on Snowflake, then merge into RTO_REGION_HOURLY_STAGING
Schedule: @daily
Next Run ID: 2025-12-07, 00:00:00 UTC
12/07/2025 06:02:08 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred
failed
queued
removed
restarting
running
scheduled
shutdown
skipped
success
up_for_reschedule
up_for_retry
upstream_failed
no_status
Duration
Dec 06, 00:00
00:01:59
00:00:59
00:00:00
show_env
dbt_run
merge_electricity_to_analytics
dbt_test
dbt_snapshot
DAG
eia_ELT_dbt
Run
2025-12-06, 00:00:00 UTC
Task
dbt_snapshot
Clear task
Mark state as...
Filter DAG by task
Details
Graph
Gantt
Code
Event Log
Logs
XCom
Task Duration
Layout:
Left -> Right
show_env
success
BashOperator
dbt_run
success
BashOperator
merge_electricity_to_analytics
success
PythonOperator
dbt_test
success
BashOperator
dbt_snapshot
success
BashOperator
React Flow

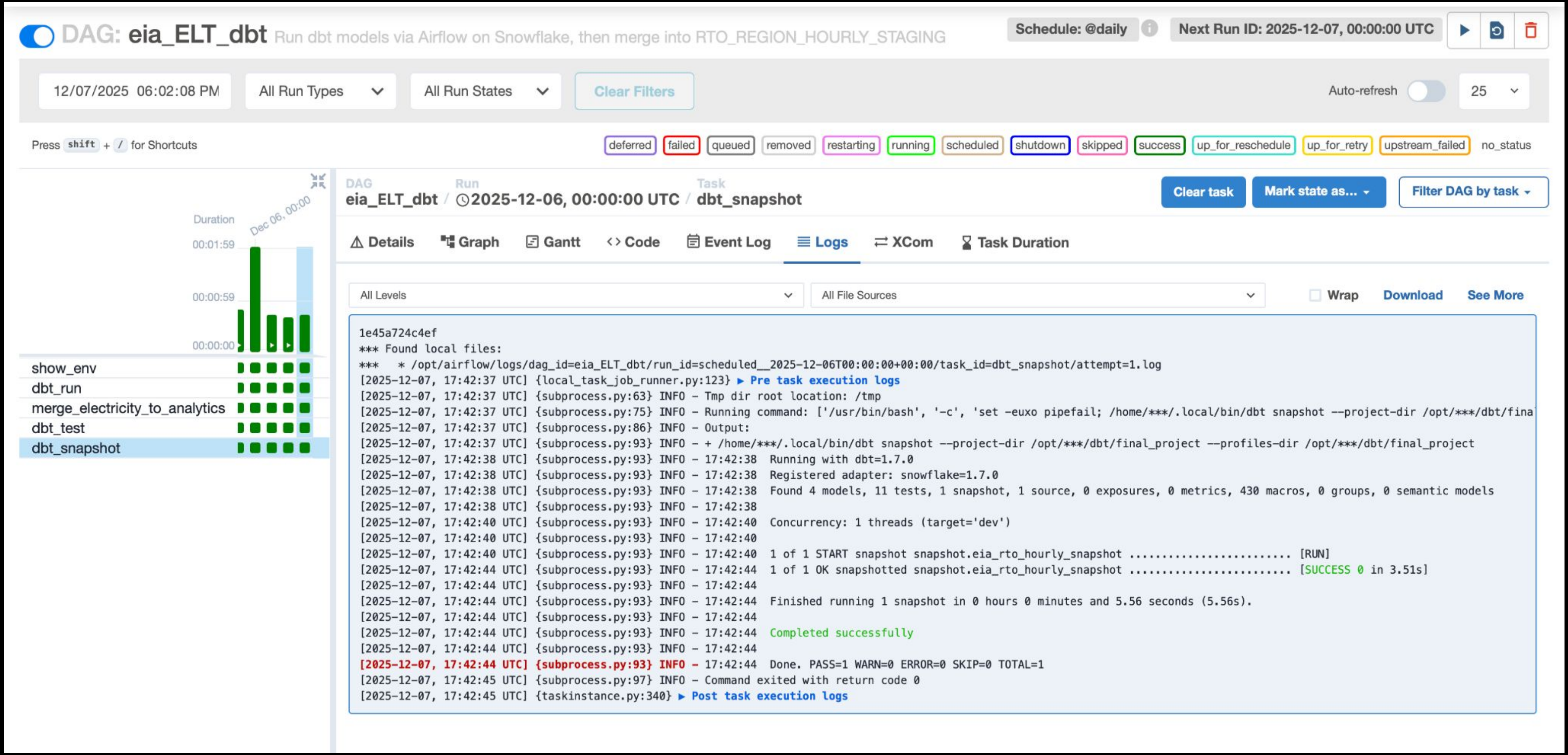
ELT (dbt run):


DAG: eia_ELT_dbt Run dbt models via Airflow on Snowflake, then merge into RTO_REGION_HOURLY_STAGING
Schedule: @daily
Next Run ID: 2025-12-07, 00:00:00 UTC
12/07/2025 06:02:08 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred failed queued removed restarting running scheduled shutdown skipped success up_for_reschedule up_for_retry upstream_failed no_status
Duration
Dec 06, 00:00
00:01:59
00:00:59
00:00:00
show_env
dbt_run
merge_electricity_to_analytics
dbt_test
dbt_snapshot
DAG eia_ELT_dbt / Run 2025-12-06, 00:00:00 UTC / Task dbt_run
Clear task
Mark state as...
Filter DAG by task
Details Graph Gantt Code Event Log Logs XCom Task Duration
All Levels
All File Sources
Wrap Download See More
1e45a724c4ef
*** Found local files:
***   * /opt/airflow/logs/dag_id=eia_ELT_dbt/run_id=scheduled__2025-12-06T00:00:00+00:00/task_id=dbt_run/attempt=1.log
[2025-12-07, 17:42:05 UTC] {local_task_job_runner.py:123} ▶ Pre task execution logs
[2025-12-07, 17:42:05 UTC] {subprocess.py:63} INFO - Tmp dir root location: /tmp
[2025-12-07, 17:42:05 UTC] {subprocess.py:75} INFO - Running command: ['/usr/bin/bash', '-c', 'set -euxo pipefail; /home/***/.local/bin/dbt run --project-dir /opt/***/dbt/final_pro
[2025-12-07, 17:42:05 UTC] {subprocess.py:86} INFO - Output:
[2025-12-07, 17:42:05 UTC] {subprocess.py:93} INFO - + /home/***/.local/bin/dbt run --project-dir /opt/***/dbt/final_project --profiles-dir /opt/***/dbt/final_project
[2025-12-07, 17:42:06 UTC] {subprocess.py:93} INFO - 17:42:06 Running with dbt=1.7.0
[2025-12-07, 17:42:07 UTC] {subprocess.py:93} INFO - 17:42:07 Registered adapter: snowflake=1.7.0
[2025-12-07, 17:42:07 UTC] {subprocess.py:93} INFO - 17:42:07 Found 4 models, 11 tests, 1 snapshot, 1 source, 0 exposures, 0 metrics, 430 macros, 0 groups, 0 semantic models
[2025-12-07, 17:42:08 UTC] {subprocess.py:93} INFO - 17:42:08
[2025-12-07, 17:42:12 UTC] {subprocess.py:93} INFO - 17:42:12 Concurrency: 1 threads (target='dev')
[2025-12-07, 17:42:12 UTC] {subprocess.py:93} INFO - 17:42:12
[2025-12-07, 17:42:12 UTC] {subprocess.py:93} INFO - 17:42:12 1 of 4 START sql view model analytics_staging.stg_eia .... [RUN]
[2025-12-07, 17:42:13 UTC] {subprocess.py:93} INFO - 17:42:13 1 of 4 OK created sql view model analytics_staging.stg_eia .... [SUCCESS 1 in 0.94s]
[2025-12-07, 17:42:13 UTC] {subprocess.py:93} INFO - 17:42:13 2 of 4 START sql table model analytics.moving_avg .... [RUN]
[2025-12-07, 17:42:15 UTC] {subprocess.py:93} INFO - 17:42:15 2 of 4 OK created sql table model analytics.moving_avg .... [SUCCESS 1 in 2.25s]
[2025-12-07, 17:42:15 UTC] {subprocess.py:93} INFO - 17:42:15 3 of 4 START sql table model analytics.region_comp .... [RUN]
[2025-12-07, 17:42:17 UTC] {subprocess.py:93} INFO - 17:42:17 3 of 4 OK created sql table model analytics.region_comp .... [SUCCESS 1 in 1.73s]
[2025-12-07, 17:42:17 UTC] {subprocess.py:93} INFO - 17:42:17 4 of 4 START sql table model analytics.weekday_weekend .... [RUN]
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18 4 of 4 OK created sql table model analytics.weekday_weekend .... [SUCCESS 1 in 1.50s]
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18 Finished running 1 view model, 3 table models in 0 hours 0 minutes and 10.57 seconds (10.57s).
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18 Completed successfully
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18
[2025-12-07, 17:42:18 UTC] {subprocess.py:93} INFO - 17:42:18 Done. PASS=4 WARN=0 ERROR=0 SKIP=0 TOTAL=4

ELT (dbt test):


DAG: eia_ELT_dbt Run dbt models via Airflow on Snowflake, then merge into RTO_REGION_HOURLY_STAGING
Schedule: @daily
Next Run ID: 2025-12-07, 00:00:00 UTC
12/07/2025 06:02:08 PM
All Run Types
All Run States
Clear Filters
Auto-refresh
25
Press shift + / for Shortcuts
deferred failed queued removed restarting running scheduled shutdown skipped success up_for_reschedule up_for_retry upstream_failed no_status
Duration
Dec 06, 00:00
00:01:59
00:00:59
00:00:00
show_env
dbt_run
merge_electricity_to_analytics
dbt_test
dbt_snapshot
DAG eia_ELT_dbt / Run 2025-12-06, 00:00:00 UTC / Task dbt_test
Clear task
Mark state as...
Filter DAG by task
Details Graph Gantt Code Event Log Logs XCom Task Duration
All Levels
All File Sources
Wrap Download See More
1e45a724c4ef
*** Found local files:
*** * /opt/airflow/logs/dag_id=eia_ELT_dbt/run_id=scheduled__2025-12-06T00:00:00+00:00/task_id=dbt_test/attempt=1.log
[2025-12-07, 17:42:23 UTC] {local_task_job_runner.py:123} ▶ Pre task execution logs
[2025-12-07, 17:42:23 UTC] {subprocess.py:63} INFO - Tmp dir root location: /tmp
[2025-12-07, 17:42:23 UTC] {subprocess.py:75} INFO - Running command: ['/usr/bin/bash', '-c', 'set -euxo pipefail; /home/***/.local/bin/dbt test --project-dir /opt/***/dbt/final_pr
[2025-12-07, 17:42:23 UTC] {subprocess.py:86} INFO - Output:
[2025-12-07, 17:42:23 UTC] {subprocess.py:93} INFO - + /home/***/.local/bin/dbt test --project-dir /opt/***/dbt/final_project --profiles-dir /opt/***/dbt/final_project
[2025-12-07, 17:42:24 UTC] {subprocess.py:93} INFO - 17:42:24 Running with dbt=1.7.0
[2025-12-07, 17:42:25 UTC] {subprocess.py:93} INFO - 17:42:25 Registered adapter: snowflake=1.7.0
[2025-12-07, 17:42:25 UTC] {subprocess.py:93} INFO - 17:42:25 Found 4 models, 11 tests, 1 snapshot, 1 source, 0 exposures, 0 metrics, 430 macros, 0 groups, 0 semantic models
[2025-12-07, 17:42:25 UTC] {subprocess.py:93} INFO - 17:42:25
[2025-12-07, 17:42:27 UTC] {subprocess.py:93} INFO - 17:42:27 Concurrency: 1 threads (target='dev')
[2025-12-07, 17:42:27 UTC] {subprocess.py:93} INFO - 17:42:27
[2025-12-07, 17:42:27 UTC] {subprocess.py:93} INFO - 17:42:27 1 of 11 START test accepted_values_stg_eia_SERIES__D__DF__NG__TI .... [RUN]
[2025-12-07, 17:42:28 UTC] {subprocess.py:93} INFO - 17:42:28 1 of 11 PASS accepted_values_stg_eia_SERIES__D__DF__NG__TI .... [PASS in 0.68s]
[2025-12-07, 17:42:28 UTC] {subprocess.py:93} INFO - 17:42:28 2 of 11 START test accepted_values_weekday_weekend_weekday_weekend__WEEKDAY__WEEKEND [RUN]
[2025-12-07, 17:42:29 UTC] {subprocess.py:93} INFO - 17:42:29 2 of 11 PASS accepted_values_weekday_weekend_weekday_weekend__WEEKDAY__WEEKEND . [PASS in 0.69s]
[2025-12-07, 17:42:29 UTC] {subprocess.py:93} INFO - 17:42:29 3 of 11 START test not_null_moving_avg_DATE .... [RUN]
[2025-12-07, 17:42:29 UTC] {subprocess.py:93} INFO - 17:42:29 3 of 11 PASS not_null_moving_avg_DATE .... [PASS in 0.82s]
[2025-12-07, 17:42:29 UTC] {subprocess.py:93} INFO - 17:42:29 4 of 11 START test not_null_moving_avg_REGION .... [RUN]
[2025-12-07, 17:42:30 UTC] {subprocess.py:93} INFO - 17:42:30 4 of 11 PASS not_null_moving_avg_REGION .... [PASS in 0.62s]
[2025-12-07, 17:42:30 UTC] {subprocess.py:93} INFO - 17:42:30 5 of 11 START test not_null_moving_avg_SERIES .... [RUN]
[2025-12-07, 17:42:31 UTC] {subprocess.py:93} INFO - 17:42:31 5 of 11 PASS not_null_moving_avg_SERIES .... [PASS in 0.57s]
[2025-12-07, 17:42:31 UTC] {subprocess.py:93} INFO - 17:42:31 6 of 11 START test not_null_moving_avg_VALUE .... [RUN]
[2025-12-07, 17:42:32 UTC] {subprocess.py:93} INFO - 17:42:32 6 of 11 PASS not_null_moving_avg_VALUE .... [PASS in 1.06s]
[2025-12-07, 17:42:32 UTC] {subprocess.py:93} INFO - 17:42:32 7 of 11 START test not_null_moving_avg_value_ma_7 .... [RUN]
[2025-12-07, 17:42:32 UTC] {subprocess.py:93} INFO - 17:42:32 7 of 11 PASS not_null_moving_avg_value_ma_7 .... [PASS in 0.53s]
[2025-12-07, 17:42:32 UTC] {subprocess.py:93} INFO - 17:42:32 8 of 11 START test not_null_stg_eia_DATE .... [RUN]
[2025-12-07, 17:42:33 UTC] {subprocess.py:93} INFO - 17:42:33 8 of 11 PASS not_null_stg_eia_DATE .... [PASS in 0.62s]
[2025-12-07, 17:42:33 UTC] {subprocess.py:93} INFO - 17:42:33 9 of 11 START test not_null_stg_eia_REGION .... [RUN]

ELT (dbt snapshot):



Visualization

This dashboard provides an interactive, high-level analysis of hourly electricity demand across major U.S. balancing authorities. Users can filter by date, hour, region, region group, and series type (such as demand, forecast, or net generation). Key metrics summarize overall system behavior.

The visualizations highlight different analytical perspectives:

1. Daily Electricity Value Trends show how selected series evolve over time.

2. Regional Group Comparison ranks geographic regions by average electricity value, enabling quick identification of high- and low-demand areas.

3. Series Distribution Donut Chart breaks down total electricity values by metric category.

4. Regional Summary Matrix compares weekday vs. weekend patterns across regions, emphasizing operational differences.

5. Hourly Heat Map reveals intra-day demand fluctuations, with darker shades representing higher electricity values.

Overall, the dashboard integrates temporal, regional, and categorical insights to support monitoring, comparison, and interpretation of electricity system performance.

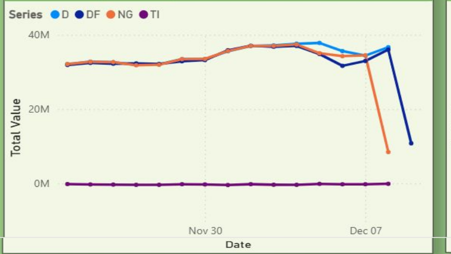
Power Pulse-Demand & Generation Analysis

This dashboard presents hourly electricity data from multiple US balancing authorities, including metrics for demand, forecast, generation, and power interchange. It enables comparison across regions, identification of temporal patterns, and evaluation of operational behavior using aggregated and time series views.
Region: The geographic balancing authority area reporting electricity data.
Series: The type of metric, such as demand, forecast, generation, or interchange.
Value: The measured electricity quantity for that region and metric.

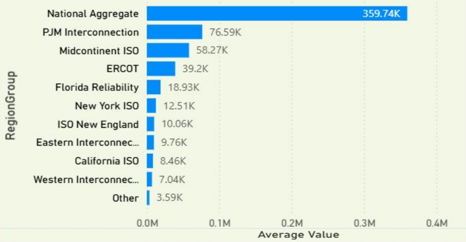
Date	Hour	Region Group	Region	Series
All	All	All	All	All

560.45K	-11.428K	15.64K	16.37K	All
Peak Value (MW)	Min Value (MW)	Average Value (MW)	Moving Avg 7 (MW)	Selected Region

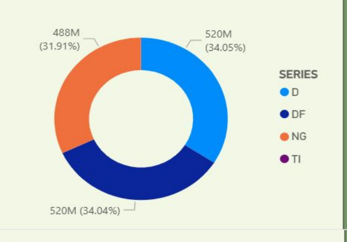
Daily Electricity Value Trends Across Series



Regional Group Comparison of Average Electricity Value



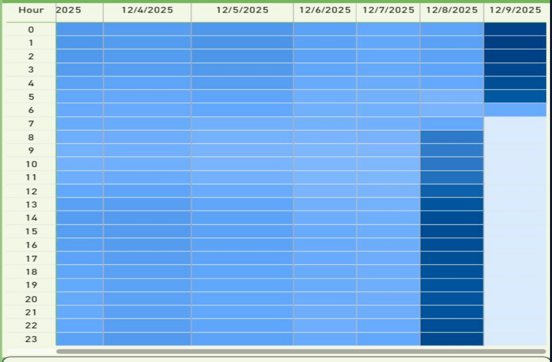
Distribution of Total Electricity Value by Series



Regional Summary Matrix

DAY	WEEKDAY			WEEKEND		
Region	Average Hourly Value	Weekday Avg	Weekend Avg	Average Hourly Value	Weekday Avg	Weekend Avg
California ISO	8,456.29	8,240.27	8,320.45	8,456.29	8,240.27	8,320.45
BANC	1,325.11	1,275.37	1,354.31	1,325.11	1,275.37	1,354.31
CISO	15,345.91	15,205.18	15,286.58	15,345.91	15,205.18	15,286.58
Eastern Interconnection	9,761.93	9,915.66	9,913.89	9,761.93	9,915.66	9,913.89
ERCOT	39,200.52	38,322.66	39,797.83	39,200.52	38,322.66	39,797.83
Florida Reliability	18,927.46	18,697.38	18,871.61	18,927.46	18,697.38	18,871.61
FLA	18,927.46	18,697.38	18,871.61	18,927.46	18,697.38	18,871.61
ISO New England	10,059.27	10,302.68	10,284.64	10,059.27	10,302.68	10,284.64
ISNE	10,059.27	10,302.68	10,284.64	10,059.27	10,302.68	10,284.64
Midcontinent ISO	58,267.98	60,138.79	59,893.98	58,267.98	60,138.79	59,893.98
MISO	58,267.98	60,138.79	59,893.98	58,267.98	60,138.79	59,893.98
National Aggregate	359,740.85	364,965.91	373,063.71	359,740.85	364,965.91	373,063.71
US48	359,740.85	364,965.91	373,063.71	359,740.85	364,965.91	373,063.71
New York ISO	12,511.14	12,566.42	12,667.89	12,511.14	12,566.42	12,667.89
NY	12,733.42	12,576.39	12,667.12	12,733.42	12,576.39	12,667.12
NYIS	12,295.48	12,556.45	12,668.65	12,295.48	12,556.45	12,668.65
Other	3,586.28	3,578.68	3,745.31	3,586.28	3,578.68	3,745.31
PJM Interconnection	76,589.13	79,141.92	79,061.74	76,589.13	79,141.92	79,061.74
Western Interconnection	7,042.09	7,120.67	7,109.40	7,042.09	7,120.67	7,109.40
Total	**15,640.85**	**15,739.58**	**15,961.46**	**15,640.85**	**15,739.58**	**15,961.46**

Hourly Electricity Value Heat Map (MW)



*Higher values shown in darker shades indicate periods of elevated electricity demand or generation.

Conclusion

Our project delivered a fully automated ETL → ELT → BI pipeline that converts hourly EIA grid data into structured analytical models and interactive dashboards, providing clear visibility into electricity behavior across U.S. regions.

Key Takeaways:

- Regions show consistent weekday–weekend demand patterns and strong evening peaks, with noticeable differences in average hourly values across authorities.
- Average hourly electricity values vary noticeably across regions, highlighting distinct load profiles and regional characteristics
- The dashboard surfaces anomalies and sudden drops, proving useful for monitoring data quality and operational irregularities.

This system creates real-world impact by giving grid operators clearer visibility into how electricity demand behaves across hours, days, and regions. By transforming raw operational data into timely, actionable analytics, it supports faster decision-making, better forecasting, and early detection of unusual load patterns. These insights help improve grid stability, reduce operational risks, and enable more efficient integration of renewable energy resources.

Summary and lessons learned

Project Summary

- Built an automated **ETL → ELT → BI** pipeline using Airflow, Snowflake, dbt, and Power BI.
- Ingested hourly data from the **U.S. EIA** to create clean, consistent, near–real-time operational datasets.
- Delivered analytical models (moving averages, regional comparisons, weekday/weekend trends) for grid behavior insights.
- Final Power BI dashboards revealed **demand patterns, regional differences, anomalies, and peak-hour trends.**
- The system supports more reliable forecasting by giving operators **accurate, timely, structured data.**

Lessons Learned

- **Airflow** ensures timely and consistent ingestion.
- **Python ETL** standardizes formats, timestamps, and regional identifiers.
- **Snowflake** provides stable, schema-consistent storage.
- **dbt** enforces clean transformations and surfaces anomalies.
- **Power BI** visually highlights patterns and irregularities.
- Good data engineering is essential for **accurate forecasting and grid reliability.**

Future work

Integrate Real-Time Data:

- Add CAISO 5–10 min interval streams for near real-time monitoring and anomaly detection.

Expand Analytical Models:

- Incorporate short-term forecasting (ARIMA, ML-based), peak prediction, and renewable variability analysis.

Improve Data Quality & Pipelines:

- Implement incremental loading, add dbt freshness checks, and expand testing coverage.

User & Dashboard Enhancements:

- Enable alerts for spikes or anomalies; build deeper forecasting and region-level exploration pages.

Scalability & Deployment:

- Scale Airflow/Snowflake for larger workloads and containerize the pipeline for CI/CD and smoother deployment.

Thank you!